Application Note

This document describes how elliptic integrals and functions are used in the R47/C47 calculator.

In some cases, elliptic function parameters are used differently depending on which book or software package you are looking at. This is a matter of slight differences in terminology and notation across the elliptic function literature. Especially notable is the modulus parameter, interpreted differently and with the letter k or m, with different meaning. When comparing computations from one place or another, the answers may appear to disagree, but only because the parameter was used differently.

This document attempts to explain those differences so that you can get consistent results. If parameters are mapped properly from one usage to another, the answers should come out well.

Numerical results are given to 5 digits to the right of the decimal point where possible, for consistency in comparisons. The text [1] uses that precision, which is easily set on the R47/C47 with `FIX 5`.

1 Basics of Elliptical Functions and Integrals

What are elliptic functions and integrals?

In some ways they are just solutions to differential equations that have no apparent bearing on ellipses at all.

Here we will explore those issues and show simple examples of computations with elliptic functions that you can try on an R47 or C47 calculator, or in Mathematica.

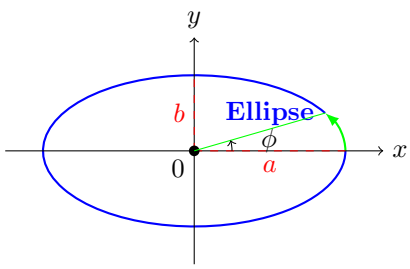
1.1 Review: Basic Geometry of an Ellipse

Many people studied conic sections in high school. See [3] for example. A conic section is a geometrical idea from ancient Greek mathematics. Certain curves can be represented as a conic section geometrically combined with analytical (Cartesian) geometry. Examples:

Since we are interested in ellipses here, let us note that the ellipse has eccentricity $e \in [0, 1)$. In the special case where an ellipse has eccentricity 0, it is actually a circle.

Table 1: Examples of Conic Sections

Name	Equation	Eccentricity, e
Circle	$x^2 + y^2 = r^2$	0
Parabola	$y^2 = 4ax$	1
Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$\sqrt{1 - \frac{b^2}{a^2}}$
Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\sqrt{1 + \frac{b^2}{a^2}}$



Besides the equation given above for an ellipse, it can also be represented parametrically:

$x = a\cos\theta, y = b\sin\theta, \theta \in [0, 2\pi)$

1.2 A Problem to Solve with an Ellipse: Arc Length and Inverse

One sees online comments, e.g. [4], about elliptic integrals and functions deriving from arc length problems, but actual examples are hard to find. Most of the applications for elliptic integrals and functions go back to problems from specialized differential equations or engineering problems.

I persisted in this question and found answers that tie back to the functions in the R47/C47 calculator. Here are my findings.

First, there is the problem of notation and parameter names. This is discussed in [5]. An elliptic function like $sn(u, m)$ can be described with either m or k given as the second argument, and sometimes the second argument is not mentioned at all.

Similarly, an elliptic integral can be described as $K(m)$ or $F(\phi, k)$ or $F(\phi, m)$, with apparently k and m being different things. And then, their relationship to geometric parameters that we know from simple conic descriptions, like a or b or e, is not obvious.

The simple answer is, that when you see k as a parameter, it is called the "modulus", but actually it is equal to the eccentricity e as seen in elementary conic section terminology. Some sources instead refer to m as the modulus.

However, though k is often favoured in mathematical treatments, you often see in software such as the R47/C47 or Mathematica a parameter called m. Actually, this is almost the same thing: $m = k^2 = e^2$. This m parameter is just the square of eccentricity.

Now, a typical second year calculus textbook for engineering students teaches how to calculate arc length in two- or three- dimensional Euclidean space, using integration. The Olmsted book, [2], does a good job of this.

To paraphrase [2]: If C is an arc with parametrization functions $x = f(t), y = g(t)$, with continuous derivatives on $[a, b]$, then C is rectifiable with length

$$L = \int_a^b \sqrt{(f'(t))^2 + g'(t))^2} dt$$

As an example of this, for the ellipse described above, we get (with a little rearrangement):

The total arc length of the ellipse is [2]

$$4a \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta$$

This leads to the following:

Incomplete elliptic integral of the second kind:

$$E(\phi, m) = \int_0^{\phi} \sqrt{1 - m \sin^2 \theta} d\theta$$

Complete elliptic integral of the second kind:

$$E(m) = \int_0^{\pi/2} \sqrt{1 - m \sin^2 \theta} d\theta$$

Confusingly, in different parts of the literature, you will see this version:

Incomplete elliptic integral of the second kind:

$$E(\phi, k) = \int_0^{\phi} \sqrt{1 - k^2 \sin^2 \theta} d\theta$$

Complete elliptic integral of the second kind:

$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta$$

See also [8].

Just understand that k is eccentricity of the ellipse, $m = k^2$, and that for both you have the restriction $k \in [0, 1), m \in [0, 1)$.

So now we have an understandable application if you followed all the bits about conic sections, eccentricity, and arc length. We simply want to take an ellipse and calculate its arc length. The incomplete elliptic integral of the second kind does this for arbitrary angle ϕ, meaning we start at angle 0 and rotate counter-clockwise to ϕ, measuring arc length as we go. The complete elliptic integral of the second kind calculates the perimeter of the ellipse.

Let us take an example and run it through both Mathematica and the R47/C47.

Take $a = 5, b = 1$, so we have the ellipse $\frac{x^2}{5^2} + \frac{y^2}{1} = 1$.

We have eccentricity $k = e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{24/25}$. Both Mathematica and the R47/C47 use a parameter named m, so we use modulus $m = .96$.

In Mathematica, we can compute this:

```
EllipticE[.96]

--> 1.05050
```

As a sanity check, we can also compute:

```
Integrate[Sqrt[1 - .96 (Sin[t])^2], {t, 0, Pi/2}]

--> 1.05050
```

Also:

```
ArcLength[{5 Sin[t], Cos[t]}, {t, 0, Pi/2}]

--> 5.25251
```

which is 5 times the value above, also confirming our correctness.

If we want to calculate the partial arc length, say for an angle ϕ of .2 radians and the same modulus, we run

```
EllipticE[.2, .96]

--> 0.19872
```

On the R47/C47, we do this:

```
.96 E(m)
```

```
--> 1.0505
```

For the partial angle:

```
.96 Enter .2 E(φ,m)

--> .19872
```

This confirms that elliptic integrals of the second kind give correct arc length calculations.

Now let us look at the inverse problem. For a given modulus and a given arc length, we want to know what angle ϕ produces that arc length.

Unfortunately, my researches show that there is no such function in either the mathematical literature or the software packages of Mathematica or R47/C47. We have to resort to other methods such as root solving. See [7] for more on this issue.

In Mathematica, try this:

```
FindRoot[EllipticE[x, .96] == .19872, {x, .5}]
--> 0.20000
```

Correct answer.

In the R47/C47, enter a program:

```
LBL EllE
MVAR 'x'
MVAR 'm'
MVAR 'u'
RCL 'm'
RCL 'x'
E(φ, m)
RCL- 'u'
END
```

If you run this program `EllE` under

```
ADV > SOLVE
```

with a modulus of .96 and a u value of .19872, you will get an answer of 0.20000.

This accounts for $E(m)$ and $E(\phi, m)$ on the R47/C47, which are the Elliptic Integrals of the Second Kind.

1.3 Elliptic Integrals of the First Kind

The integrals of the first kind are $K(m)$ (complete) and $F(\phi, m)$ (incomplete).

Equations:

$$F(\phi, m) = \int_0^{\phi} \frac{d\theta}{\sqrt{1 - m\sin^2\theta}}$$

$$K(m) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - m\sin^2\theta}}$$

These also have a geometric interpretation, from Complex Analysis. On constructing a conformal map from an ellipse to a unit circle, the elliptic integral of the first kind tells you the angular displacement needed to map arc length in the ellipse to the corresponding point in the circle [6].

In this case, there is an inverse mapping. $am(x)$, described in Section 1.5, is the inverse of the incomplete elliptic integral of the first kind.

1.4 Elliptic Integrals of the Third Kind and Jacobi Zeta

$\Pi(n, m)$ is the incomplete Elliptic Integral of the third kind. It has applications in Physics, e.g. electrostatic potential.

In Mathematica, use $EllipticPi[n, m]$. On the R47/C47, use $\Pi(n, m)$.

Comparison of values from Mathemetica and the R47/C47.

```
EllipticPi[0.4,0.6]

--> 2.59092
```

```
.6 ENTER .4 Π(n,m)

--> 2.59092
```

The Jacobi Zeta function is found in Mathematica as $JacobiZeta[\phi, m)$ and on the R47/C47 as $Z(\phi, m)$. It has applications in physics and in interpretation of elliptic arc compared with circle.

Mathematica example:

```
JacobiZeta[2,.5]

--> -0.11777
```

R47/C47 Example:

```
.5 ENTER 2 Z(φ,m)

--> -0.11777
```

These two match.

1.5 Jacobi Elliptic Functions

As a foundation to these functions, there are first the four Theta functions. These arise as solutions to a partial differential equation related to heat diffusion over a slab:

$$\kappa \nabla^2 \theta = \frac{d\theta}{dt}$$

These give rise to functions $\theta_1(z,q), \theta_2(z,q), \theta_3(z,q), \theta_4(z,q)$.

The R47/C47 calculator does not implement these Theta functions, but Mathematica does: $EllipticTheta[a,u,q]$. These are used for the theory behind the following Elliptic functions, which also have applications in the Elliptic integrals.

$sn(u,m), cn(u,m), dn(u,m)$. These are rendered in Mathematica as

```
JacobiSN[u,m]
JacobiCN[u,m]
JacobiDN[u,m]
```

These functions have some interesting identities. For example:

$sn(u,m)^2 + cn(u,m)^2 = 1$

$sn(u,0) = \sin(u)$ for u in radians.

$cn(u,0) = \cos(u)$ for u in radians.

We can test these. For instance, on the R47/C47:

```
0 ENTER .2 sn(u,m) .2 sin
```

puts matching values (0.19867) in the X and Y registers.

As an example of understanding how to use parameters, the book [1] shows in Table C that for $k = 0.5$ and $x = 0.8$, $sn(x,k) = 0.70421$.

However, this calculation on the R47/C47:

```
.5 ENTER .8 sn(u,m)
```

produces the number .69093: an apparent mismatch.

Is this wrong? NO. The calculator got a different answer from the book, but both the calculator and the book are right. WHAT?

Yes, explaining this issue clearly is one of the main motivations for writing this document. The answer is in the difference in parameter interpretation. At the risk of being too verbose, I **must** explain this point.

Earlier in this document, I used a lot of space to explain the k parameter versus the m parameter. Some books or software packages expect k as the modulus parameter, and some expect m as the modulus parameter. As explained earlier, $m = k^2$.

Where the book's table has $sn(u, k)$ with $k = 0.5$, the calculator wants to run $sn(u, m)$ and we have to square the k parameter to get the m parameter.

Knowing that $m = k^2$, we can recalculate on the R47/C47 this way:

`.5` x^2 `.8 sn(u,m)`

and get the matching value of 0.70421.

If this sounds strange, think of trig functions. The result of the sine function depends on whether the parameter is interpreted as being in degrees or radians or other units. Accordingly, you might have to do a conversion between degrees and radians in some cases to make the sine function give the expected result.

Here, the result of $sn()$ depends on whether the second parameter is interpreted as the eccentricity or the square of the eccentricity of the ellipse. Accordingly, you might have to square a parameter to make it compatible with the function you are calling.

More examples comparing $sn()$ calculations with Tables in the back of the [1] textbook:

Table 2: Lawden Comparison Examples: Jacobi Functions

k	x	sn x	cn x	dn x	sn(u,m)	cn(u,m)	dn(u,m)
.5	.8	.70421	.70999	.93596	.70421	.70999	.93596
.6	1.3	.93416	.35686	.82816	.93416	.35686	.82816

Incomplete Elliptic integrals using degrees:

Table 3: Lawden Comparison Examples: Incomplete Elliptic Integrals

k	$\phi°$	Lawden $F(\phi, k)$	$F(\phi, k)$	Lawden $D(\phi, k)$	$E(\phi, k)$
.4	36	.63459	.63459	.62215	.62215
.6	65	1.21431	1.21431	1.06311	1.06311

Apart from this issue for $sn()$, there is also the Jacobi amplitude function, represented in the R47/C47 as $am(u, m)$ and in Mathematica as $JacobiAmplitude[u, m]$. Here, m is the modulus representing the square of eccentricity, as before.

This function is the inverse of the elliptic integral of the first kind. To test this on the R47/C47, try an example of $\phi = .1, m = .7$. Calculation of elliptic integral of first kind:

```
.7 x² .1 F(φ,m)

--> .10008
```

Calculation of inverse with same modulus:

```
.7 x² .10008 am(u,m)

--> .10000
```

1.6 Summary of R47/C47 Elliptic Functions

The following R47/C47 functions have been described:

Table 4: R47/C47 functions described under Ellipt

R47/C47 Name	Description
$am(u, m)$	Jacobi Amplitude Function
$sn(u, m)$	Jacobi sn Function
$cn(u, m)$	Jacobi cn Function
$dn(u, m)$	Jacobi dn Function
$K(m)$	Complete Elliptic Integral of First Kind
$F(\phi, m)$	Incomplete Elliptic Integral of First Kind
$E(m)$	Complete Elliptic Integral of Second Kind
$E(\phi, m)$	Incomplete Elliptic Integral of Second Kind
$\Pi(n, m)$	Elliptic Integral of Third Kind
$Z(\phi, m)$	Jacobi Zeta

In each case except for $\Pi(n, m)$, the u or ϕ parameter is the primary argument, and the m parameter is the square of eccentricity of the ellipse. For $\Pi(n, m)$, n is the characteristic and m is the square of eccentricity.

References

[1] Lawden, Derek F, *Elliptic Functions and Applications*, Springer-Verlag, 1989

[2] Olmsted, John M H, *Advanced Calculus*, Prentice-Hall, 1961

[3] Del Grande and Egsgard, *Relations*, Gage Educational Publishing, 1972

[4] StackExchange: *What even are elliptic functions?*. `https://math.stackexchange.com/questions/3383769/what-even-are-elliptic-functions` (Accessed: April 2026).

[5] Cook, J. D., *Clearing up the confusion around Jacobi functions* `https://www.johndcook.com/blog/2018/10/12/jacobi-function-nomenclature/`

[6] StackExchange: *Elliptic integral of the first kind as conformal mapping*. `https://math.stackexchange.com/questions/2341813/elliptic-integral-of-the-first-kind-as-conformal-mapping` (Accessed: April 2026)

[7] StackExchange: *Inverse of elliptic integral of second kind*. `https://math.stackexchange.com/questions/1123360/inverse-of-elliptic-integral-of-second-kind` (Accessed: April 2026)

[8] Wikipedia: Elliptic integral `https://en.wikipedia.org/wiki/Elliptic_integral` (Accessed: April 2026)

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ANNEX 1: Tutorial to Appnote 23: calculating with ellipses

Below, complimentary to the AN0023 by Will, follows:

1. Explanatory sketches clarifying the variable used by the c47/R47 implementations,
2. Explanatory notes relating to the sketches,
3. Two worked examples, complete with detailed diagrams,
4. 20 problems with solutions, verified on C47/R47, excerpted from Baker, 1890 (see footnote 1 below).

Let me quote Baker from 1890 (see footnote 1 below):

> *It is in the hope of smoothing the road to this interesting and increasingly important branch of Mathematics, and of putting within reach of the English student a tolerably complete outline of the subject, clothed in simple mathematical language and methods, that the present work has been compiled.*
>
> *New or original methods of treatment are not to be looked for. The most that can be expected will be the simplifying of methods and the reduction of them to such as will be intelligible to the average student of Higher Mathematics.*

Funny that in 1890 already, this work was considered mature enough to not want new fashioned treatments ;-) 130 years later the R47/C47 implements still the same math with no new treatments.

R47/C47 X.FN menu ➔ R47/C47 Elliptic / Jacobi functions:

Ellipt	Orthog	$J_y(x)$	$Y_y(x)$	erf	erfc
W_m	W_p	W^{-1}	I_{xyz}	$\beta(x,y)$	LNβ
γ_{xy}	Γ_{xy}	$I\Gamma_p$	$I\Gamma_q$	$\Gamma(x)$	LNΓ

θ→m	m→θ	a,b→m	→DEG	→D.MS	→RAD
k→m	m→k	am(u,m)	F(ϕ,m)	E(ϕ,m)	Z(ϕ,m)
sn(u,m)	cn(u,m)	dn(u,m)	K(m)	E(m)	Π(n,m)

Fig. 1: The worked examples below can be replicated with the R47/C47 Ellipt menu functions above:

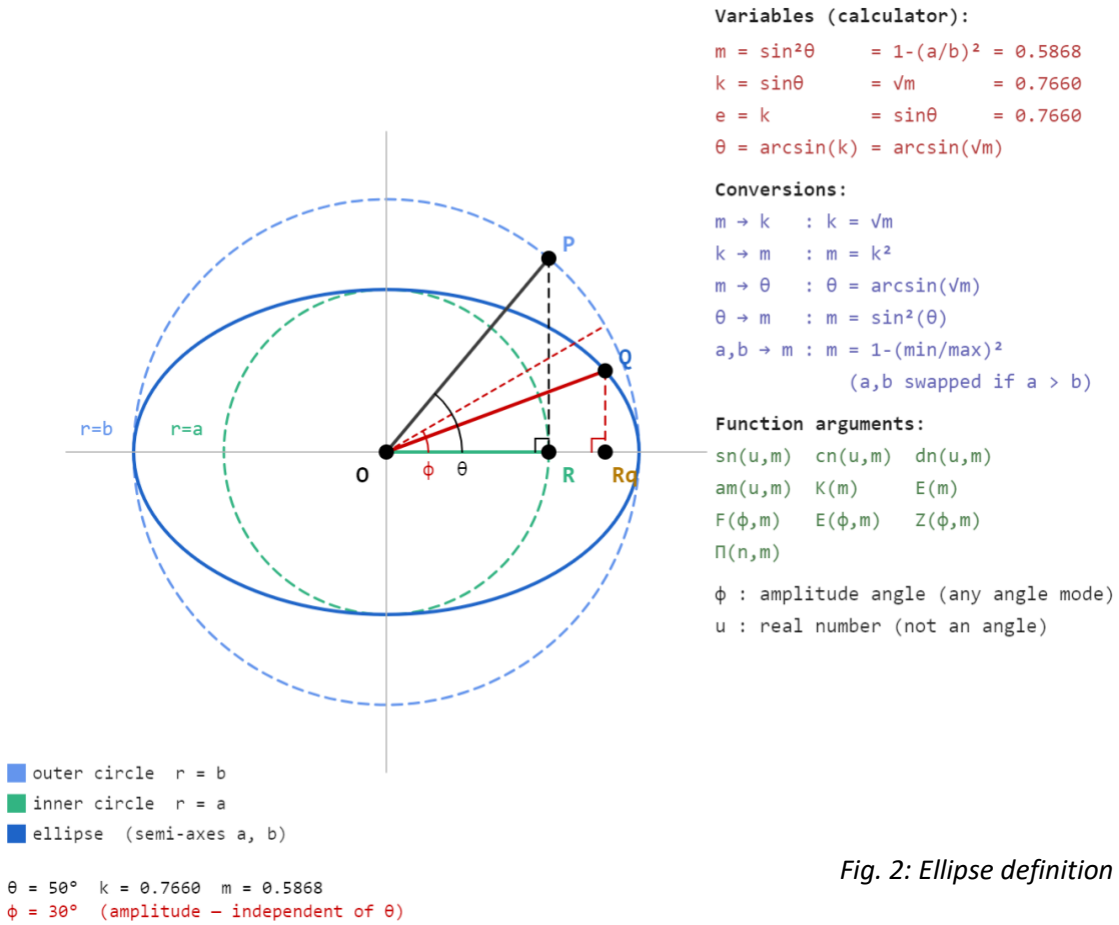

Fig. 2: Ellipse definition & command summary

The ellipse above is fully described by its two semi-axes a and b. From these the calculator parameter $m = 1-(a/b)^2$ and modulus $k = \sqrt{m}$ follow directly.

The eccentricity angle $\theta = \arcsin(k)$ ties the ellipse shape to the function parameters. With P on the outer circle at angle θ, the vertical from P drops to R on the major axis and OR = a, giving $\cos\theta = a/b$. If a and b are known, $m = 1-(a/b)^2$ follows directly. If the ellipse is given as a drawing, θ is the angle to measure to get m without needing to know a and b explicitly.

For a point Q on the ellipse, the amplitude angle ϕ gives the coordinates of Q through the ellipse equations $x = b\cdot\cos\phi$ and $y = a\cdot\sin\phi$. This is the critical distinction the diagram is meant to convey: ϕ is not the geometric angle QOR visible in the figure, which would be arctan(y/x) evaluated at Q.

The two different angles θ and ϕ, at the same point, are most easily confused.

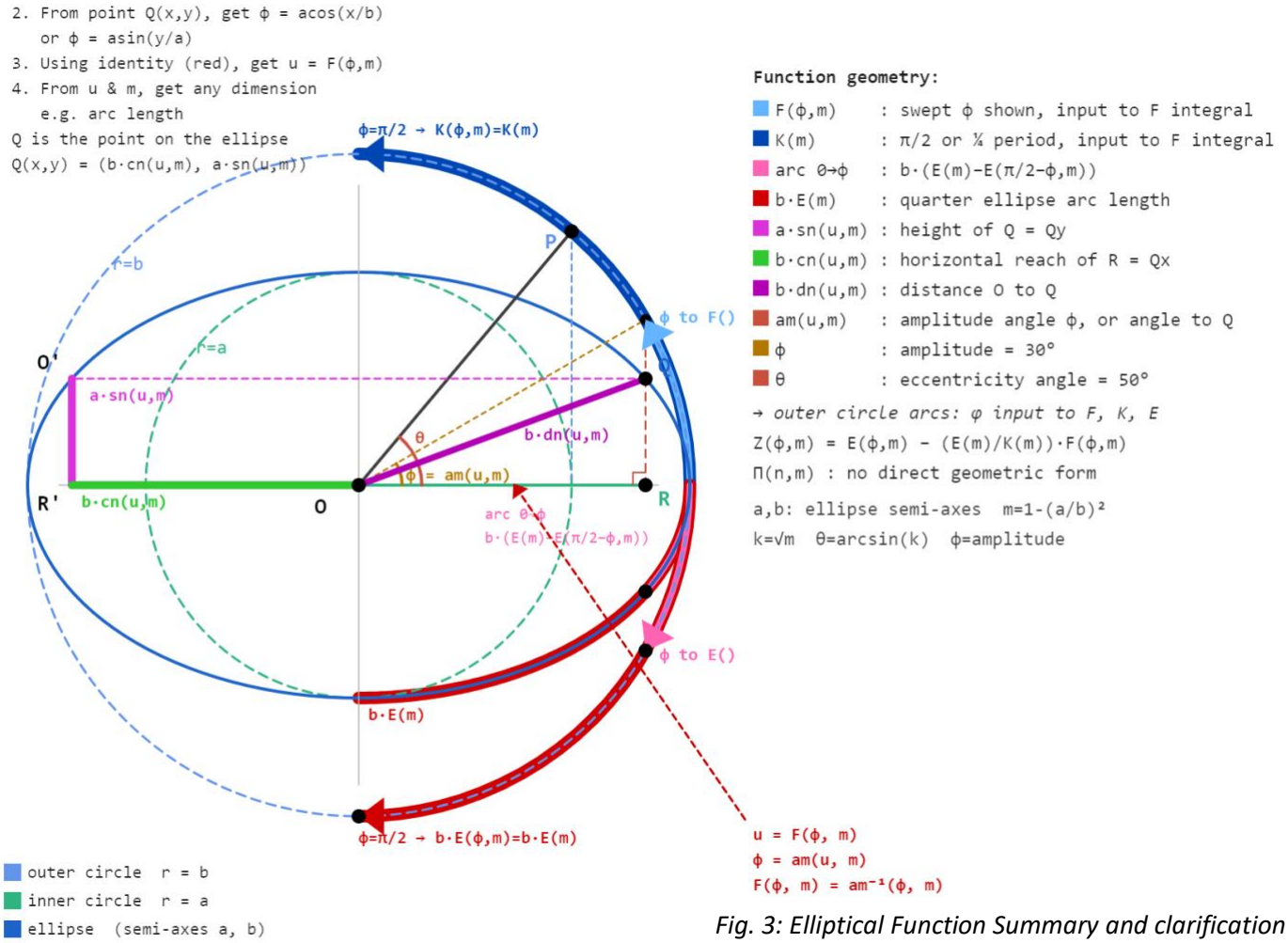

Fig. 3: Elliptical Function Summary and clarification

Once m and ϕ are known, u = F(ϕ,m) is the single quantity that unlocks all Jacobi functions. F is the inverse of the amplitude function: ϕ = am(u,m), so u is simply the argument whose amplitude is ϕ. The diagram shows this geometrically – the blue arc sweeps ϕ as input to F, and K(m) = F(π/2,m) is the same function evaluated at its quarter-period input.

The arc length along the ellipse from 0 to Q is b·(E(m) − E(π/2−ϕ,m)), not b·E(ϕ,m). The distinction matters in practice: E(ϕ,m) alone does not give arc length except at the quarter period where it equals b·E(m) exactly. The correction term E(π/2−ϕ,m) accounts for the asymmetry of the integrand between the two ends of the quarter arc.

The segments b·cn(u,m), a·sn(u,m) and b·dn(u,m) form a right triangle OQR, so once u and m are known all three dimensions of Q are available simultaneously.

A single call to F gives u, and from u the full geometry of the ellipse follows in one step.

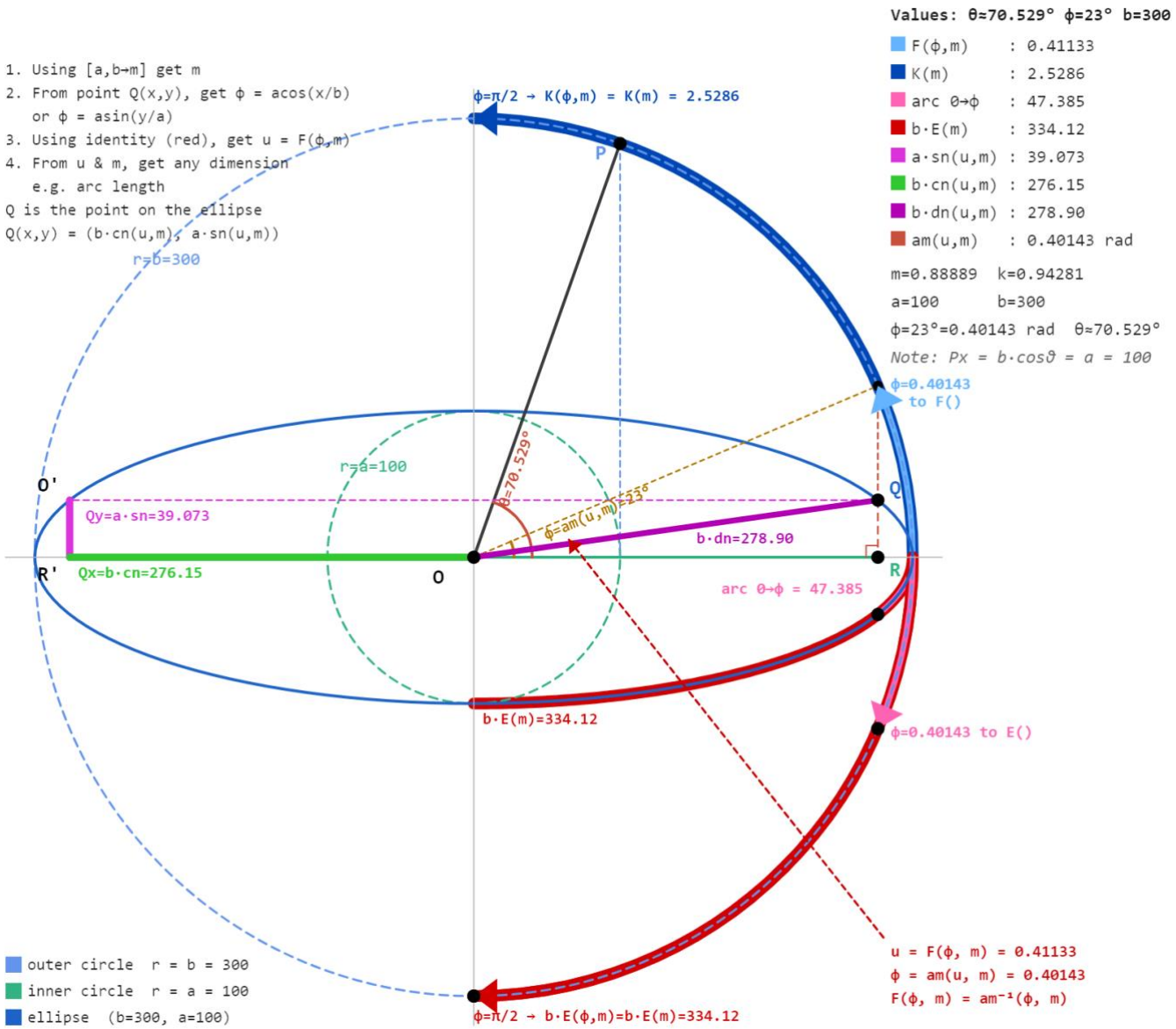

Fig. 4: Elliptical Numerical Solutions 1

Above, the highly eccentric ellipse with a=100 and b=300 which gives m=0.889 and k=0.943. At this eccentricity the elliptic functions behave quite differently from a near-circular case, and the geometry is more visible precisely because the ellipse is so elongated.

The instruction block top left gives the workflow in four steps: get m from a and b, get ϕ from the coordinates of Q, get u from F(ϕ,m), then read off any dimension. Everything flows from m and ϕ.

The blue arc shows ϕ=23° being fed into F. The output u=0.41133 is neither an angle nor a length and does not appear on the drawing. It is the argument to all the Jacobi functions. From u and m the calculator returns b·cn=276.15, a·sn=39.07 and b·dn=278.90 as the three sides of the right triangle around the point on the ellipse, i.e. OQRq. K(m)=2.529 is F evaluated at ϕ=π/2, the quarter period input.

The arc OQ along the ellipse is 47.385, obtained as b·(E(m)−E(67°,m)) where 67°=90°−ϕ. The full quarter arc b·E(m)=334.12 is considerably longer than a less eccentric ellipse would give, reflecting how much the curve stretches near the major axis when k approaches 1.

The red identity block (bottom right) states u=F(ϕ,m) and ϕ=am(u,m) as inverses.

To solve the ellipse, starting with a coordinate, arc length, or angle, requires finding ϕ, then u.

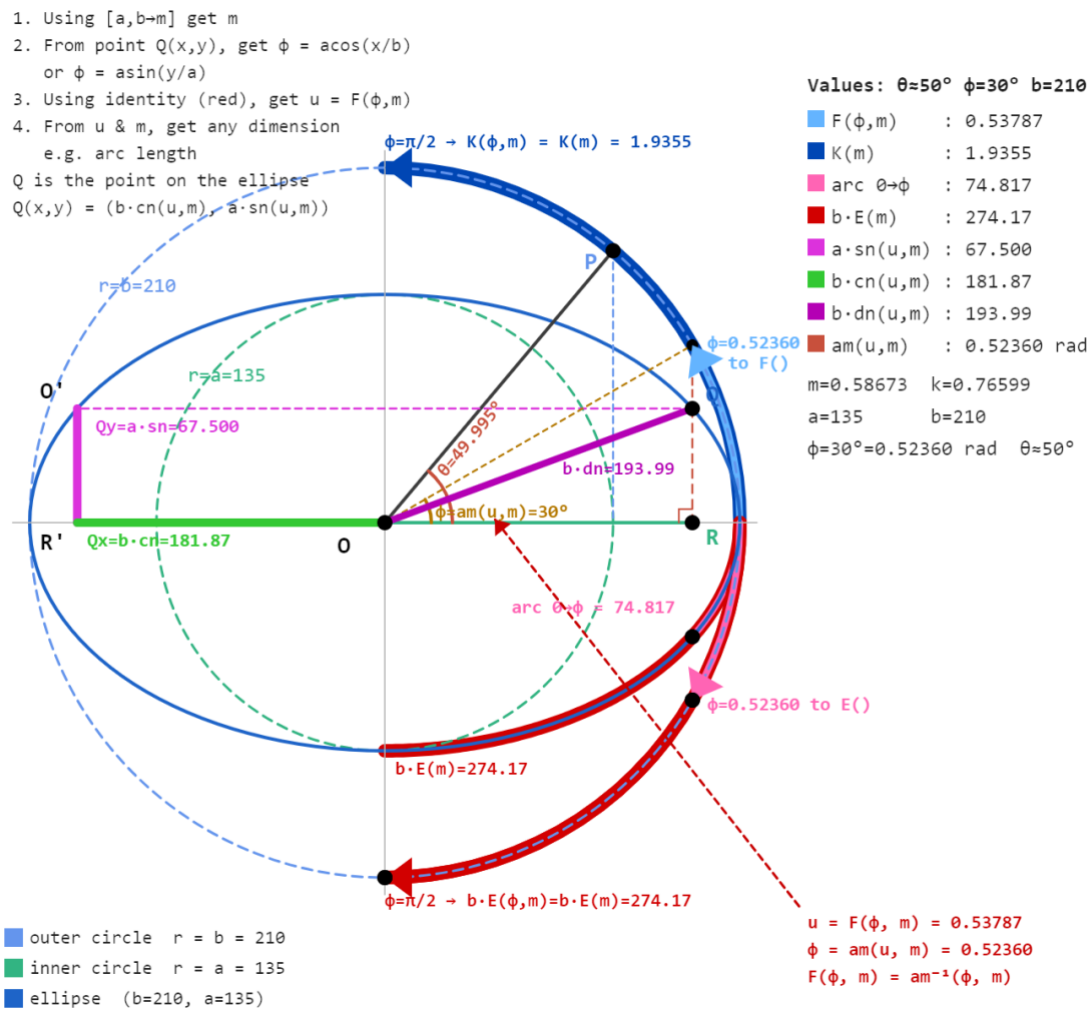

Fig. 5: Elliptical Numerical Solutions 2

Second exercise left to user, with no comments

Examples from Baker[1],

1. p.106: Let k= sin θ= sin 19°30′. Required K. K= 1.615101 (*ed. Printing error. 3 digits correct in the book; log K was given correctly as log K=0.208852, implying their log-book lookup was wrong*).

2. p.108: Given k= sin 75°. Find K. K= 2.768064 Ans.

3. p.109: Given k= sin 45°. Find K. K= 1.8540747 Ans.

4. p.110: Given θ= 63°30′. Find K. Ans. log K= 0.3539686

5. p.110: Given θ= 34°30′. Find K. Ans. K= 1.72627

6. p.111: Let ϕ = 30° k= sin 45°. Find u. u =0.535623, Ans.

7. P.113: Given k= sin 75°, tan ϕ = √(2/√3). To find F(k, ϕ). F(k, ϕ) = 0.9226878. Ans.

8. p.117: Given ϕ = 30°, k= sin 89°. Find u. F(k, ϕ) = 0.549297. Ans.

9. p.118: Given = 79°, k = 0.25882. Find u. Ans. u = 0.39947. (*ed. Printing error: 1.39947*).

10. p.118: Given = 37°, k = 0.86603. Find u. Ans. u =0.68141.

11. p.119: Given u = 1.368407, θ = 38°. Find ϕ. ϕ = 72°. Ans.

12. p.122: Given u = 2.41569, θ = 80°. Find ϕ. Ans. ϕ = 82°

13. p.122: Given u = 1.62530, k = 1/2 . Find ϕ. Ans. ϕ = 87°

14. p.123: Given k= 0.9327, ϕ = 80°. Find E(k, ϕ). E(k, ϕ) = 1.0562 (*Ed. Incorrect, it is 1.06349*)

15. p.123: Given k= sin 75°, tan ϕ = √(2/√3). To find E(k, ϕ). E(k, ϕ) = 0.7387196. Ans.

16. p.127: Given k = sin 75°. Find E(k, π). E(k, π) = 1.076405. Ans.

17. p.127: Given k =sin 30°, ϕ = 81°. Find E(k, 0). Ans. E(k, 0) =1.33124.

18. p.127: Find E(sin 80°, ϕ = 55°). Ans. 0.82417.

19. p.127: Find E(sin 27°, ϕ = π /2). Ans. 1.48642.

20. p.127: Find E(sin 19°, ϕ = 27°). Ans. 0.46946.

[1] *Baker Arthur L, Elliptic Functions An Elementary Text-Book for Students of Mathematics, New York John Wiley and Sons, 1890, re-released Project Gutenberg, January 25, 2010 [eBook #31076]*

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